

Coffee Shop Analysis

Overview

A Simple, but very insightful analysis that revolves around product participation throughout April-September 2025. The goal was to obtain trends, patterns, and propose financial feedback for future

Only provided with monthly items and could only perform revenue analysis. The future plan is to gain access to the cost of goods to calculate marginal monthly profits.

Dataset

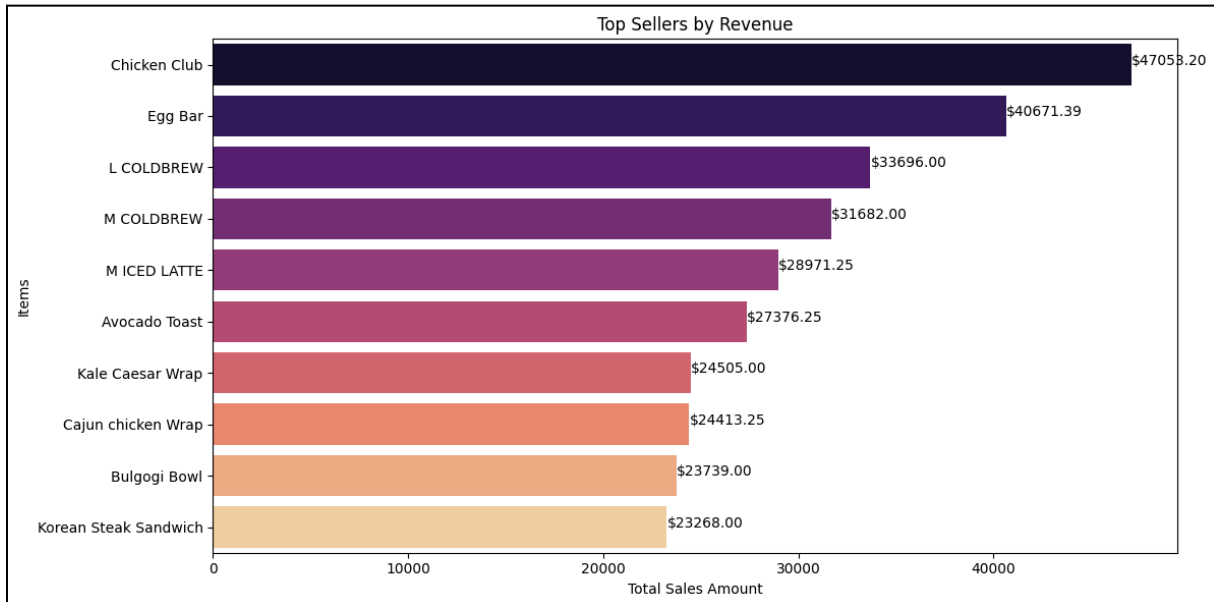
- Monthly data from April to September of 2025 from a local coffee shop in Jersey City.
- Manually cleaned individual datasets from miscellaneous rows/columns using Excel and Python.
- Aggregated all month's dataset into one 6_month_sales dataset, later using this to perform revenue analysis.

Methods

1. Cleaned all datasets from each month for better use.
 - a. Extracted the top 10 most sold items by *sales amount* and *quantity*.
 - b. Utilized *ExponentialSmoothing* from *statsmodel* library to perform a 4 month revenue forecast using “multiplicative” as the trend with a seasonal_period of 2 (since I don't have a full cycle of season yet given the limited amount of data provided).
2. Aggregated all months into one big dataset to report a 6 month report:
 - a. Extracted the top 10 most sold items by *sales amount* and *quantity*.
3. Displayed insightful charts to encapsulate monthly performance, top 10 items sold by sales amount and quantity, and a 4 month revenue forecast.

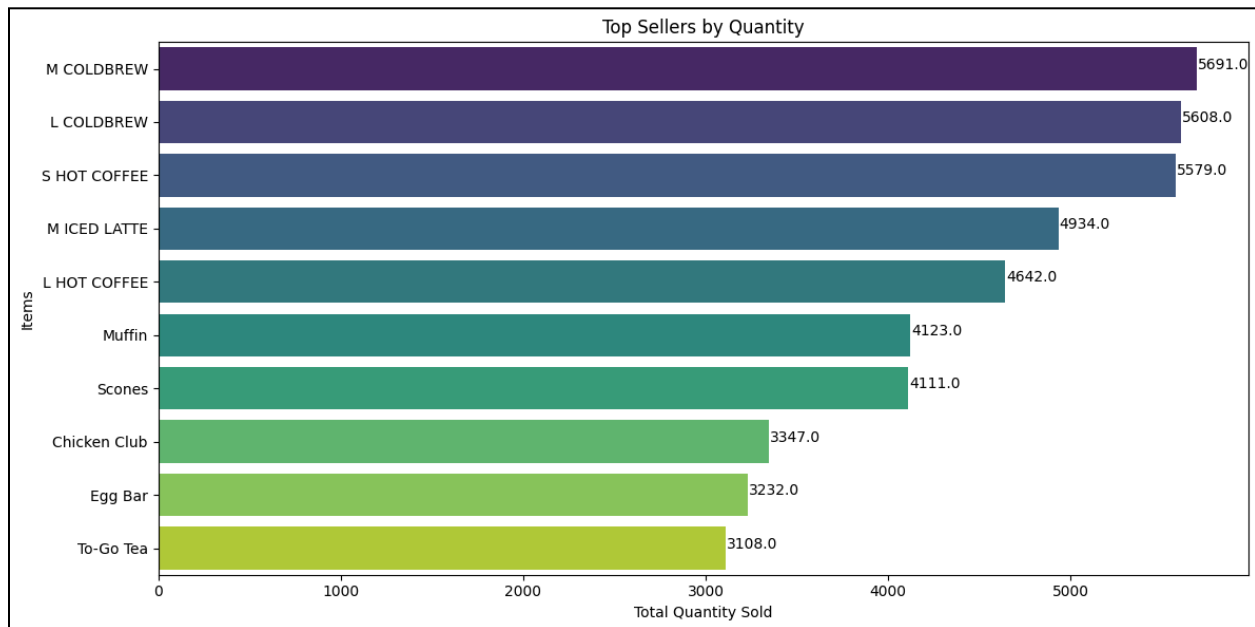
Learnings & Insights

Figure 1



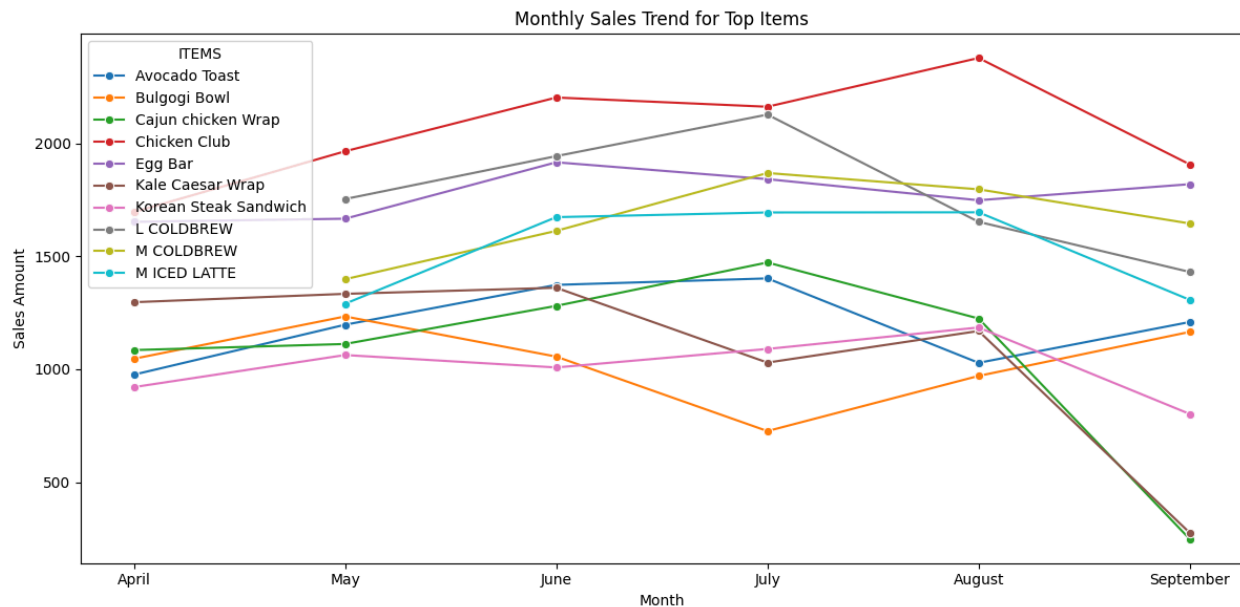
- For the 6 month period, chicken club is the apparent customer favorite with a ~\$7,000 lead ahead of egg bar.
- Recommendations to further push greater revenue in the future:
 1. Increase the price of *Chicken Club* by 7-8%: Not too high that it starts driving away customers from the product, let alone the store, but enough to keep them paying for something they're willing to pay (quality-price trade-off).
 2. **Promote *Egg Bar***: It falls behind Chicken Club by only ~\$7,000
 - a. Have waiters/waitresses promote the egg bar to customers firsthand.

Figure 2



- It makes sense that the top 5 items are coffee products.
- Medium size seems to be the go to by customers.
 - Although *M COLDBREW* is the top seller by quantity, it does *NOT* bring in the most money.
- Going back to promoting the *egg bar* since it appears in both top seller charts; promoting this item can both bring in more money and quantity sold.
 - Idea: having a promotion with the *M COLDBREW* can serve as a campaign to sell both items at a discounted price, but still gain positive revenue.

Figure 3



- There's a steady incline of each top 10 items from April through June (rolling in to the summer)
 - More people are outside and motivated to go out and
- Then, from August to September, there's a drastic decline in quantity sales from 6 items.
 - This can stem from the fall season and customers are less motivated to order their usuals.
 - Better promote seasonal menus, especially drinks as a different outlet of revenue from other items in decline.

Overall, despite only using such a short period of data, I was able to gain insight on trends and patterns in a real-life business. With my recommendations, I can further back it up with statistical models/predictions, but with such limited data, such predictions become too linear, underfitting the overall performance of the coffee shop.